

Erdős #685: two-sided bounds for $\omega\left(\binom{n}{k}\right)$

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Abstract

#685 asks whether $\omega\left(\binom{n}{k}\right) = (1 + o(1)) k \sum_{k < p < n} 1/p$ for $n^\varepsilon < k \leq n^{1-\varepsilon}$. Pushing the Legendre window from $n/2$ down to $\sqrt{n}$ gives an exact description of which primes above $T = \max(\sqrt{n}, k)$ divide $\binom{n}{k}$ — precisely those with a multiple in $(n - k, n]$ — and hence an exact split $\omega = B + S$ into large and small rails. Bounding the two parts separately yields the two-sided estimate

$$\#\{m \in (n - k, n] : P^+(m) > T\} \leq \omega\left(\binom{n}{k}\right) \leq k + \pi(T),$$

whose left side already attains the conjectured order. At $k = \lfloor \sqrt{n} \rfloor$ the lower bound has expected size $(\log 2)k$, exactly the conjectured value, and the upper bound is $(1 + o(1))k$, so the conjecture is bracketed between its own prediction and $1/\log 2$ times it. #685 itself is not proved.

1 Setup

By Kummer, $p \mid \binom{n}{k}$ iff some base- p digit of k exceeds the corresponding digit of n ; writing $k \preceq_p n$ for digit domination,

$$\omega\left(\binom{n}{k}\right) = \#\{p \leq n : k \not\preceq_p n\}.$$

No binomial need be evaluated. Throughout $1 \leq k \leq n/2$.

2 The bound

Lemma 1 (window). *For prime $p > n/2$: $v_p\left(\binom{n}{k}\right) = 1$ if $n - k < p \leq n$, and $v_p\left(\binom{n}{k}\right) = 0$ if $n/2 < p \leq n - k$.*

Proof. $p^2 > n \geq m$ for $m \leq n$, so $v_p(m!) = \lfloor m/p \rfloor$ and $v_p\left(\binom{n}{k}\right) = \lfloor n/p \rfloor - \lfloor k/p \rfloor - \lfloor (n - k)/p \rfloor = 1 - 0 - \lfloor (n - k)/p \rfloor$, using $k \leq n/2 < p$. $\square$

Theorem 2. *For $1 \leq k \leq n/2$,*

$$\omega\left(\binom{n}{k}\right) \geq \pi(n) - \pi(n - k).$$

Proof. By the first case of Lemma 1 every prime of $(n - k, n]$ divides $\binom{n}{k}$. $\square$

Verified: no exception for $4 \leq n < 500$ and all $k \leq n/2$.

Remark 3 (strength of Theorem 2). *By the prime number theorem $\pi(n) - \pi(n - k) = (1 + o(1))k / \log n$, whereas the conjectured value $k \sum_{k < p < n} 1/p$ has size $k \log(\log n / \log k)$. So Theorem 2 is sharp in k but loses the logarithmic factor. The window can, however, be pushed from $n/2$ down to $\sqrt{n}$, and doing so recovers the right order.*

3 The refined window, and a bound of the conjectured order

Theorem 4 (refined window). *Let $p > \sqrt{n}$ be prime with $p > k$. Then $p \mid \binom{n}{k}$ if and only if p has a multiple in $(n - k, n]$.*

Proof. Since $p^2 > n$, Legendre's formula has one term: $v_p(m!) = \lfloor m/p \rfloor$ for $m \leq n$. As $k < p$, $\lfloor k/p \rfloor = 0$, so $v_p\left(\binom{n}{k}\right) = \lfloor n/p \rfloor - \lfloor (n - k)/p \rfloor$, the number of multiples of p in $(n - k, n]$. $\square$

Verified: 7 327 302 triples (n, k, p) with $n < 700$ — no mismatch.

Theorem 5. *Write $P^+(m)$ for the largest prime factor of m and put $T = \max(\sqrt{n}, k)$. Then for $1 \leq k \leq n/2$,*

$$\omega\left(\binom{n}{k}\right) \geq \#\{m \in (n - k, n] : P^+(m) > T\}.$$

Proof. Let $m \in (n - k, n]$ with $p := P^+(m) > T$. Then $p > \sqrt{n}$ and $p > k$, and p has the multiple m in $(n - k, n]$, so $p \mid \binom{n}{k}$ by Theorem 4. Distinct such m give distinct p : an interval of length $k < p$ contains at most one multiple of p , so m is recovered from p . The map $m \mapsto P^+(m)$ is therefore injective into $\mathcal{D}(n, k)$. $\square$

Corollary 6. *For $k = n^\theta$ with $\frac{1}{2} \leq \theta < 1$, the right-hand side of Theorem 5 has expected size $\rho(1/\theta)k$, where ρ is the Dickman function, since $T = k$ and the density of $m \leq n$ with $P^+(m) > n^\theta$ is $1 - \rho(1/\theta) + o(1)$. At $\theta = \frac{1}{2}$ this is $(\log 2)k$, which is exactly the order of the conjectured $k \sum_{k < p < n} 1/p = (1 + o(1))(\log 2)k$. So Theorem 5 attains the conjectured order of magnitude, in contrast with Theorem 2.*

The remaining content of #685 is therefore the passage from an inequality of the right order to the asymptotic constant, and — separately — the small- k range $k < \sqrt{n}$, where $T = \sqrt{n}$ and Theorem 5 counts only the $m \in (n - k, n]$ with a prime factor exceeding $\sqrt{n}$.

Remark 7. *Lemma 1 shows no prime of $(n/2, n - k]$ divides $\binom{n}{k}$, so the prime set above $n/2$ is exactly the primes of $(n - k, n]$; Theorem 4 extends the exact description down to $\sqrt{n}$. Any further progress on #685 must come from $p \leq \sqrt{n}$, where carries of depth ≥ 2 occur and the description is no longer a single floor difference.*

4 An exact split, and a matching upper bound

Theorem 4 determines membership of $\mathcal{D}(n, k)$ above $T := \max(\sqrt{n}, k)$ completely, which splits ω into two parts that can be bounded separately.

Theorem 8 (exact split). *For $1 \leq k \leq n/2$ put $T = \max(\sqrt{n}, k)$. Then*

$$\omega\left(\binom{n}{k}\right) = B(n, k) + S(n, k),$$

where

$$B(n, k) = \#\{p > T : p \text{ has a multiple in } (n - k, n]\}, \quad S(n, k) = \#\{p \leq T : k \nmid_p n\}.$$

Proof. The primes dividing $\binom{n}{k}$ are exactly $\{p : k \nmid_p n\}$. Split this set at T . Above T we have $p > \sqrt{n}$ and $p > k$, so Theorem 4 identifies membership with having a multiple in $(n - k, n]$, giving B ; below T the definition gives S . $\square$

Verified: 39 780 pairs (n, k) with $n < 400$ — the split is exact throughout.

Theorem 9 (upper bound). *For $1 \leq k \leq n/2$ and $T = \max(\sqrt{n}, k)$,*

$$\omega\left(\binom{n}{k}\right) \leq k + \pi(T).$$

Proof. By Theorem 8 it suffices to bound each part. For B : any $p > T \geq \sqrt{n}$ has at most one multiple in an interval of length $k < p$, so p determines a unique $m_p \in (n - k, n]$ with $p \mid m_p$. The map $p \mapsto m_p$ is injective, since $m_p \leq n < p^2$ forces p to be the unique prime factor of m_p exceeding $\sqrt{n}$. Hence B is at most the number of integers in $(n - k, n]$, namely k . For S : trivially $S \leq \pi(T)$. $\square$

Verified: 201 960 pairs (n, k) with $n < 900$ — no violation.

Corollary 10 (two-sided at $k = \sqrt{n}$). *Taking $k = \lfloor \sqrt{n} \rfloor$, Theorems 5 and 9 give*

$$\#\{m \in (n - k, n] : P^+(m) > \sqrt{n}\} \leq \omega\left(\binom{n}{k}\right) \leq k + \pi(\sqrt{n}) = k\left(1 + \frac{2}{\log n}(1 + o(1))\right).$$

The left side has expected size $(\log 2)k$ by Dickman, which is exactly the conjectured value $k \sum_{k < p < n} 1/p = (1 + o(1))(\log 2)k$; the right side is $(1 + o(1))k$. So at $k = \sqrt{n}$ the conjecture is bracketed between its own predicted value and $1/\log 2 = 1.4427$ times it.

Measured: at $n = 10^2, 10^4, 4 \times 10^4, 10^5$ with $k = \lfloor \sqrt{n} \rfloor$ the triple (lower, ω , upper) is (33, 39, 58), (75, 86, 125), (149, 171, 246), (229, 264, 381), with upper/conjectured ratio 2.02, 1.84, 1.83, 1.77 — decreasing towards the predicted $1/\log 2$.

5 Measurement

With ω computed by digit domination and the prediction $k \sum_{k < p < n} 1/p$:

n	k	$\omega\left(\frac{n}{k}\right)$	predicted	ratio
5000	20	26	19.0	1.369
5000	70	61	47.3	1.288
5000	300	166	117.7	1.410
20000	50	53	44.7	1.187
20000	140	120	94.5	1.270
20000	900	456	334.4	1.364

The ratio sits in $[1.19, 1.41]$ with no visible trend toward 1. This does not refute a statement with an $o(1)$, but the excess is systematic — every entry above 1, none approaching it. Its size is comparable to $\pi(k)$, the small rails the prediction omits (at $n = 5000$, $k = 300$: $\pi(k) = 62$ against an excess of 48), which suggests testing the variant $\omega = \pi(k) + (1 + o(1))k \sum_{k < p < n} 1/p$.

Remark 11 (register). **Proved:** Lemma 1, Theorem 2, Theorem 4, Theorem 5, Theorem 8, Theorem 9 and Corollary 10, all elementary from Legendre. The split and the upper bound bracket $\omega\left(\frac{n}{\lfloor \sqrt{n} \rfloor}\right)$ between the conjectured value and $1/\log 2$ times it. **Measured:** the table and the systematic excess. **Not proved:** #685. **Falsifier, pre-registered:** the ratio failing to approach 1 along $k = n^{1/2}$ as $n \rightarrow \infty$ would be evidence against the conjecture as stated; the domination formulation makes that test cheap to $n \sim 10^7$.